

Homework 1

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Exercise 1:

Properties of sequences (40pts) Let $\{\vec{x}_n\}$ and $\{\vec{y}_n\}$ be two sequences of points in $\mathbb{R}^m$, and let $\lambda \in \mathbb{R}$ be a real number. Suppose that $\{\vec{x}_n\}$ converges to a point $\vec{p}$, and $\{\vec{y}_n\}$ converges to a point $\vec{q}$. By giving a formal $\varepsilon - N$ proof, establish the following:

(10pts) The sequence $\{\lambda \vec{x}_n\}$ converges to $\lambda \vec{p}$, Hint: consider $\lambda = 0$ as a separate case.

(10pts) The sequence $\{\vec{z}_n = \vec{x}_n + \vec{y}_n\}$ converges to $\vec{p} + \vec{q}$. (20pts) The sequence of real numbers $\{a_n = \langle \vec{x}_n, \vec{y}_n \rangle\}$ given by the inner product of $\vec{x}_n$ with $\vec{y}_n$ converges to the inner product $\langle \vec{p}, \vec{q} \rangle$. Hint: Using the following, apply the Cauchy-Schwarz inequality

$$\langle \vec{x}, \vec{y} \rangle - \langle \vec{p}, \vec{q} \rangle = \langle \vec{x} - \vec{p}, \vec{y} - \vec{q} \rangle + \langle \vec{p}, \vec{y} - \vec{q} \rangle + \langle \vec{x} - \vec{p}, \vec{q} \rangle$$

Since $\vec{x}_n$ converges to $\vec{p} \rightarrow \exists \epsilon > 0$ s.t

$$|\vec{x}_n - \vec{p}| < \epsilon.$$

let $\lambda \in \mathbb{R}$

$$\begin{aligned} \exists |\lambda| \neq 0 &\implies |\lambda| (|\vec{x}_n - \vec{p}|) < |\lambda|\epsilon. \\ &\implies |\lambda\vec{x}_n - \lambda\vec{p}| < \lambda\epsilon. \end{aligned}$$

Now we consider the case $\lambda = 0$, Then $\forall \vec{x}_k \in \{\vec{x}_n\}$ we have $\lambda\vec{x}_k$ where $\lambda = 0 \implies 0\vec{x}_k = \vec{0}$. Thus if $\lambda = 0$ $\{\vec{x}_n\} = \vec{0}$.

We have the for $\{\vec{x}_n\}$ converges to $\vec{p}$. Thus by the same reasoning $\{\vec{y}_n\}$ converges to $\vec{q}$. Thus $\forall k \in \mathbb{N}$, $\vec{x}_k \rightarrow \vec{p}$, $\vec{y}_k \rightarrow \vec{q}$. Then if we pick any $\vec{y}_k + \vec{x}_k \rightarrow \vec{q} + \vec{p} = \{\vec{z}_n\}$

Cauchy swartz ineq:

$$| \langle \vec{x}, \vec{y} \rangle | \leq \|\vec{x}\| \|\vec{y}\|.$$

$$\langle \vec{x}, \vec{y} \rangle - \langle \vec{p}, \vec{q} \rangle = \langle \vec{x} - \vec{p}, \vec{y} - \vec{q} \rangle + \langle \vec{p}, \vec{y} - \vec{q} \rangle + \langle \vec{x} - \vec{p}, \vec{q} \rangle.$$

Taking the magnitude:

$$|\langle \vec{x}, \vec{y} \rangle - \langle \vec{p}, \vec{q} \rangle| = |\langle \vec{x} - \vec{p}, \vec{y} - \vec{q} \rangle + \langle \vec{p}, \vec{y} - \vec{q} \rangle + \langle \vec{x} - \vec{p}, \vec{q} \rangle|$$

.

$$\begin{aligned} |\langle \vec{x}, \vec{y} \rangle - \langle \vec{p}, \vec{q} \rangle| &= |\langle \vec{x} - \vec{p}, \vec{y} - \vec{q} \rangle + \langle \vec{p}, \vec{y} - \vec{q} \rangle + \langle \vec{x} - \vec{p}, \vec{q} \rangle| \\ |\langle \vec{x} - \vec{p}, \vec{y} - \vec{q} \rangle + \langle \vec{p}, \vec{y} - \vec{q} \rangle + \langle \vec{x} - \vec{p}, \vec{q} \rangle| &\leq |\langle \vec{x} - \vec{p}, \vec{y} - \vec{q} \rangle| + |\langle \vec{p}, \vec{y} - \vec{q} \rangle| + |\langle \vec{x} - \vec{p}, \vec{q} \rangle|. \end{aligned}$$

$$\leq \|\vec{x} - \vec{p}\| \|\vec{y} - \vec{q}\| + \|\vec{p}\| \|\vec{y} - \vec{q}\| + \|\vec{x} - \vec{p}\| \|\vec{q}\|.$$

We see each term converges as $\vec{x}, \vec{y}$ converge and $\vec{p}, \vec{q}$ are finite. Thus the three terms converge to some finite α, β, γ respectively. Thus we have:

$$|\langle \vec{x}, \vec{y} \rangle - \langle \vec{p}, \vec{q} \rangle| \leq \alpha + \beta + \gamma = \epsilon.$$

So $\langle \vec{x}, \vec{y} \rangle - \langle \vec{p}, \vec{q} \rangle =$ converges.

$$\implies \langle \vec{x}, \vec{y} \rangle \rightarrow \langle \vec{p}, \vec{q} \rangle.$$

Exercise 2 Matrix norms (60pts) In the following, you may use any standard facts from your Let $A : \mathbb{R}^m \rightarrow \mathbb{R}^s$ be a linear transformation represented by the $(s \times m)$ -matrix $(A_{i,j})_{\substack{1 \leq i \leq s \\ 1 \leq j \leq m}}$ with respect to the standard bases. We define the Hilbert-Schmidt norm of T by

$$\|A\|_{HS} := \sqrt{\sum_{i=1}^s \sum_{j=1}^m A_{i,j}^2}$$

(15 pts) Show that, for any $\vec{x} \in \mathbb{R}^m$,

$$\|A\vec{x}\| \leq \|A\|_{HS} \|\vec{x}\|$$

Hint: what are the components of $A\vec{x}$ and how could we bound them using the Cauchy-Schwarz inequality? (15 pts) Show that, given linear transformations

$$A : \mathbb{R}^m \rightarrow \mathbb{R}^s \text{ and } B : \mathbb{R}^s \rightarrow \mathbb{R}^t$$

the Hilbert Schmidt norm satisfies

$$\|BA\|_{HS} \leq \|B\|_{HS} \|A\|_{HS}$$

(15 pts) Denoting by A^T the transpose of the matrix A , and by $\text{tr}(M)$ the trace of a square matrix M , show that

$$\|A\|_{HS}^2 = \text{tr}(A^T A)$$

(15 pts) Let $A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be given by the below matrix, and define a sequence of points in $\mathbb{R}^3$ by $\vec{x}_n := A^{n-1} \vec{x}_1$, where $\vec{x}_1$ is given below. Prove that $\{\vec{x}_n\}$ converges to $\vec{0}$.

$$A = \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{5} & \frac{1}{10} & 0 \end{pmatrix}, \quad \vec{x}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

2a

$$(A\vec{x})_i = \sum_{n=1}^m A_{i,j} x_j.$$

$$\|A\vec{x}\| = \sqrt{\sum_{n=i}^s (Ax)_i^2}.$$

Cauchy swartz implies:

$$| \langle A_{i,j}, x_j \rangle |^2 \leq \|A_{i,j}\|^2 \|x_i\|^2.$$

$$\|A\vec{x}\| \leq \sqrt{\sum_{n=i}^s \left(\sum_{j=1}^m A_{i,j} \right)^2} \sqrt{\sum_{j=1}^m x_j^2}.$$

$$\|A\vec{x}\| \leq \|A\|_{HS} \|\vec{x}\|.$$

2b

We have

$$\|BA\|_{HS} = \sqrt{\sum_{i=1}^t \left(\sum_{q=1}^m (BA)_k^2 \right)}.$$

$$| \langle B_{j,k}, A_{j,i} \rangle | \leq \|A_i\| \|B_k\|$$

$$\|BA\|_{HS} \leq \sqrt{\sum_{k=1}^t \left(\sum_{j=1}^s B_j^2 \right)} \sqrt{\sum_{i=1}^m \left(\sum_{j=1}^m A_j^2 \right)}.$$

$$\|BA\|_{HS} \leq \|B\|_{HS} \|A\|_{HS}.$$

2c

$$\begin{aligned}\left(A^T A\right)_{i,k} &= \sum_{j=1}^s A_{j,i} A_{j,k}. \\ \text{tr}\left(A^T A\right) &= \sum_{i=1}^m \left(A^T A\right)_{i,i}. \\ &= \sum_{i=1}^m \sum_{j=1}^s A_{i,j}^2.\end{aligned}$$

And we have

$\|A\|_{HS}^2 = \sum_{i=1}^s \sum_{j=1}^m A_{i,j}^2$. Where i and j are just bookkeeping indices. Thus we have

$$\|A\|_{HS}^2 = \text{tr}\left(A^T A\right).$$

as required.

2d

Here I calculated the eigenvalues using mathematica to be (approximately).

$\vec{x}_1 = \begin{pmatrix} 2.879 \\ 0.979 \\ 1 \end{pmatrix}, \vec{x}_2 = \begin{pmatrix} -3.669 \\ 10.974 \\ 1 \end{pmatrix}, \vec{x}_3 = \begin{pmatrix} -0.0710 \\ -0.0620 \\ 1 \end{pmatrix}$, Thus we can diagonalise the matrix to get $A = SJS^{-1}$. Explicitly we have:

$$A = \begin{pmatrix} -0.710 & -3.669 & 2.879 \\ -0.620 & 10.974 & 0.979 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} -0.710 & 0 & 0 \\ 0 & 0.364 & 0 \\ 0 & 0 & .674 \end{pmatrix} \begin{pmatrix} -0.216 & -0.141 & 0.759 \\ -0.035 & 0.077 & 0.024 \\ 0.250 & 0.064 & 0.674 \end{pmatrix}.$$

Now we can use the properties of diagonalised matrices. ie $A^k = SJ^kS^{-1}, \forall k > 0$. Here we see that we are raising the J matrix which consists of a diagonal with all entries ≤ 1 . We know that $\forall p \in \mathbb{R} ||p|| \leq 1$ as we take p^k for large k $p^k \rightarrow 0$. Thus we are multiplying each component by smaller and smaller numbers J_i , each $J_i \leq 1 \leq \epsilon$ for some finite ϵ . Since $\vec{x}_n$ is finite and we multiply by a matrix which converges to $\vec{0}$ then the composition of $\vec{x}_n = A^{n-1}\vec{x}_1$ must converge to $\vec{0}$.